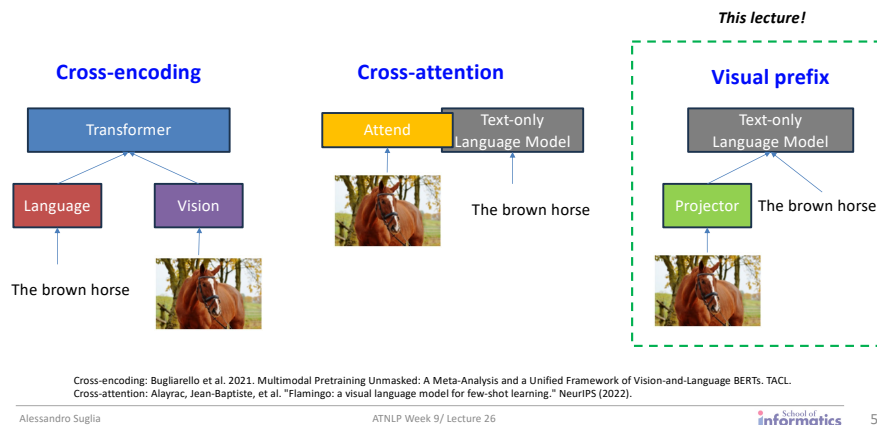


Multimodal Generative AI Models: Designing and Implementing Vision and Language Models

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

课程定位：从上一讲的 grounding 动机，进入如何设计、实现和训练现代 Vision+Language Models → 架构选择：cross-encoding、cross-attention、visual prefix 三条路线中，本讲聚焦 visual prefix → 核心组件：language expert = text-only LLM, vision expert = pretrained ViT / contrastive encoder, projector 负责把视觉空间接到语言空间 → 视觉 token 化：ViT 把图像切成 patches，分辨率越高或 patch 越小，visual tokens 越多，显存和上下文成本越高 → 视觉编码器预训练：vision-only reconstruction 能学视觉结构，vision+language contrastive encoders 如 CLIP/SigLIP 更天然对齐语言 → 训练目标：visual prefix VLM 通常用 next-token cross-entropy，但 loss 只算 language tokens，visual tokens 只是条件上下文 → 实现案例：SmolVLM 用 SigLIP + SmolLM2 + projector，是 open weight、可复现、面向小显存部署的代表 → 效率机制：image splitting、video frame sampling 和 pixel shuffle 共同处理高分辨率、多图、视频带来的 token 压力 → 训练课程：adaptation stage 学 projector，image stage 强化 OCR/文档/图表，video stage 扩展到多图和视频 → 评测与研究：VLMEvalKit、VRAM/性能权衡、context length、vision encoder size、mechanistic interpretability 和 object-centric representations

Main Approaches for Designing VLMs



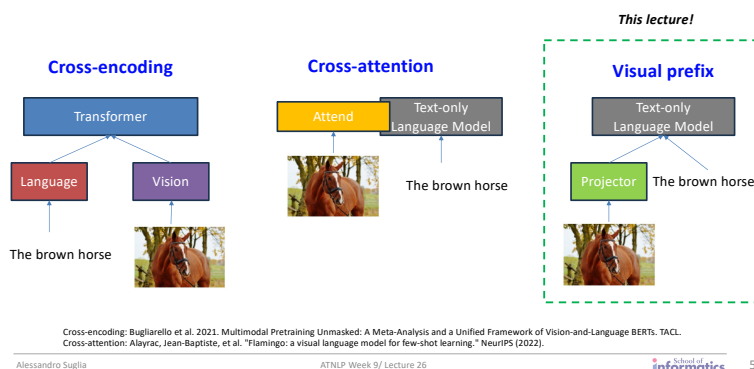
原 PPT 第 5 页

这节课一句话

这节课一句话：frontier VLM 的主流做法不是从零训练一个“既会看又会说”的怪物，而是把预训练 vision expert 看成眼睛、预训练 LLM 看成大脑，再用 projector 把图像变成 visual prefix tokens；真正难点在于视觉 token 数量、跨模态对齐、冻结/微调策略、数据课程和效率之间做工程级权衡。

1. Visual prefix: 把图像翻译成 LLM 能读的前缀

Main Approaches for Designing VLMs



原 PPT 第 5 页

人话直觉

先把三种 VLM 架构想成三种“图文开会”方式。Cross-encoding 是让图片 token 和文字 token 一起坐进同一个会议室；cross-attention 是文字代表需要时转头问视觉代表；visual prefix 最像给 LLM 递一张已经翻译好的小纸条：纸条上不是自然语言，而是一串视觉 embeddings，但它们占据 prefix 位置，告诉 LLM“接下来你说话时要以这张图为上下文”。

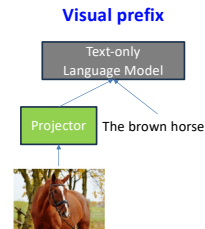
精确定义 / 机制： Visual prefix-based VLM 的核心思想是：不重新发明语言模型，而是复用强大的 text-only LLM backbone；不让 LLM 直接处理原始像素，而是先由 pretrained vision encoder 把图像变成 visual tokens；再用 projector 把这些视觉表示映射到 LLM 的 hidden size 和表示空间；最后把 projected visual tokens 作为 prefix 放进 LLM 的输入序列，让文本生成在视觉上下文条件下进行。它流行的原因是工程上很划算：LLM 已经会语言生成和推理，vision encoder 已经会看图，训练重点就变成跨模态对齐和任务适配。

考试问法：如果考“what is a visual prefix VLM”，标准答案要包含四个词：pretrained vision encoder、projector、text-only LLM、visual prefix tokens。不要只说“把图像输入 LLM”；要说视觉 encoder 输出先被 projector 映射到 LLM embedding/hidden space，然后作为 prefix conditioning language generation。

2. 组件分工：眼睛、大脑和接口层各司其职

Visual prefix-based VLMs – The intuition

- Assume that you have access to two modality experts:
 - The **language expert** = LLM
 - The **vision expert** = Pretrained Vision Transformer
- Use the pre-trained LLM as the backbone architecture
- Design a **projector module** that converts the representation of the vision expert so that they are compatible with the LLM
- Train the language model to generate tokens via **cross-entropy loss**
 - The same loss we use for training text-only LLMs



原 PPT 第 8 页

人话直觉

这页可以当成整节课的电路图：vision expert 是眼睛，负责把图片看成向量；language expert 是大脑，负责把上下文变成流畅答案；projector 是转接头，如果没有它，眼睛的插头和大脑的插座尺寸不一样，信号再强也接不上。

精确定义 / 机制： Visual prefix VLM 假设我们有两个 modality experts。Language expert 通常是 pre-trained LLM，用作生成、指令跟随和推理 backbone。Vision expert 通常是 pretrained Vision Transformer 或 vision-language contrastive encoder，用固定数量的 visual tokens 表示图像。Projector 的任务是把 vision encoder 的输出维度和表示分布转换到 LLM 可接受的 hidden size 与语义空间。实现上，projector 可以是 feedforward neural network，也可以是 cross-attention with learnable queries 等更复杂结构。训练时模型像 text-only LLM 一样生成文本 token，常用 next-token prediction 的 cross-entropy loss。

考试问法： 如果考“components of visual prefix architecture”，按职责答最清楚：vision encoder extracts visual tokens；projector maps them to the LLM hidden space；LLM consumes them as prefix context and generates language；image-text data provides captions/answers/instructions as supervision。可补充冻结策略：早期可 freeze LLM 和 vision encoder，只训练 projector，后续再部分或全模型 finetune。

3. Patch 数量：视觉 token 是 VLM 的隐形账单

Number of patches in Pretrained Visual Encoders

The Vision Transformer always extract non-overlapping square patches of size $P \times P$ from an image of size $W \times H$. This will be the “tokens” that will be processed by the VLM! Let's assume a square image and compute the number of patches:

Image Size (H×H)	Patch Size (P×P)	Number of Patches ($NP=(H/P)^2$)
224 × 224	16 × 16	$(224/16)^2 = 14^2 = 196$
224 × 224	32 × 32	$(224/32)^2 = 7^2 = 49$
224 × 224	8 × 8	$(224/8)^2 = 28^2 = 784$
384 × 384	16 × 16	$(384/16)^2 = 24^2 = 576$

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

12

原 PPT 第 12 页

人话直觉

语言模型的 context window 像一张桌子，文字 token 已经要占位置；图像一来，不是一张图片占一个位置，而是切成很多小方块，每个方块都要坐一个座位。224×224、16×16 patch 看起来很小，其实一张图就是 196 个视觉座位；如果 patch 变成 8×8，座位直接涨到 784。

精确定义 / 机制： Vision Transformer 会从尺寸为 $W \times H$ 的图像中抽取 non-overlapping square patches，每个 $P \times P$ patch 形成一个 visual token。对方形图像，patch 数量是 $\lfloor (H/P)^2 \rfloor$ 。因此 224×224 图像配 16×16 patch 得到 $14 \times 14 = 196$ patches；配 32×32 patch 得到 49；配 8×8 patch 得到 784；384×384 配 16×16 patch 得到 576。这个数字重要，是因为 visual tokens 要进入 VLM 序列：token 越多，attention 计算、显存、context 占用和融合成本越高。VLM 的“看得更细”往往不是免费的，它会用更多 token 预算买空间细节。

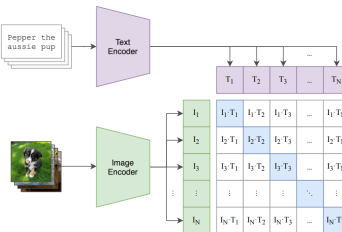
$$NP = \left(\frac{H}{P} \right)^2$$

考试问法：如果考 patch 公式，先写 $\lfloor (H/P)^2 \rfloor$ ，再做一个例子，例如 $224/16=14$ ，所以 196 patches。最后一定解释 trade-off: higher resolution 或 smaller patch size 提升细粒度视觉信息，但增加 memory/context cost；这就是后面 pixel shuffle 等压缩机制出现的原因。

4. 视觉专家：为什么 CLIP/SigLIP 比纯视觉预训练更顺手

Common Pretrained Visual Contrastive Encoders

- Combines a vision encoder with a language model
- It is trained with image-caption pairs datasets
- The objective is a contrastive loss that forces the model to bring **similar embeddings close** and **dissimilar ones far apart**
- Several SOTA variants:
 - OpenAI CLIP
 - Google SigLIP



Zhai, Xiaohua, et al. "Sigmoid loss for language image pre-training." Proceedings of the IEEE/CVF international conference on computer vision. 2023.
Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PmlR, 2021.

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

14

原 PPT 第 14 页

人话直觉

如果 vision encoder 只做过纯视觉作业，它可能很会分辨边缘、纹理、形状，但未必知道这些视觉结构怎样对应语言里的“a brown horse”。Contrastive image-language pretraining 像提前上过双语课：它已经练过把图片和 caption 拉到同一个语义附近。

精确定义 / 机制： Vision encoder 可以用 vision-only objective 预训练，例如 reconstruction loss，让模型重构被遮蔽或压缩的视觉信息；也可以用 vision+language objective 预训练，例如 contrastive loss，在 image-caption pairs 上拉近匹配图文 embedding、推远不匹配 embedding。CLIP 和 SigLIP 都属于这类视觉-语言对齐的 encoder。讲义强调，V+L data 训练出来的视觉模型通常比 vision-only models 更适合作为 VLM 的 vision expert，因为它们的表示已经接近语言语义，projector 学到的跨模态映射更容易。SmolVLM 使用 SigLIP 系列视觉编码器，正是这一思路的实际体现。

考试问法： 如果考“why use contrastive encoders in VLMs”，回答：它们用 image-caption pairs 学到 transferable visual representations and image-language alignment；相比纯视觉 reconstruction，它们更容易接入 LLM，因为视觉 embedding 已经带语言语义。可以点名 CLIP 和 SigLIP。

5. Loss 只算文字：visual tokens 是条件，不是要预测的答案

The cross-entropy loss computation for VLMs

Optimise loss

- We compute the cross-entropy loss only on the language tokens just like in a traditional LLM
- The ground truth values are typically the text tokens associated with the vision modality (e.g., caption)
- Predictions made for the visual tokens are discarded!
 - Why? We only have supervision for language tokens and none for visual tokens!

The diagram illustrates the VLM architecture. At the bottom, an image of a horse is processed by a 'Projector' (green box) to generate 'Visual tokens' (represented by grey bars). These visual tokens are fed into a 'Text-only Language Model' (grey box). Simultaneously, the caption 'The brown horse' is tokenized into 'Language tokens' (represented by grey bars labeled $L_1, L_2, L_3, \dots, L_n$). A red 'X' is placed over the visual tokens, indicating that the loss is not calculated for them. A yellow box at the top right states 'Optimise loss', indicating that the loss is calculated only on the language tokens.

Alessandro Suglia ATNLP Week 9/ Lecture 26 School of Informatics 17

原 PPT 第 17 页

人话直觉

训练 visual prefix VLM 时，图像 token 像考题里的配图，文字 token 才是答题纸。老师会批改你写出的 caption、answer 或 instruction response，但不会要求你预测“下一个视觉 token 应该是什么”，因为数据里通常没有这种视觉 token 标签。

精确定义 / 机制： Visual prefix VLM 通常沿用 LLM 的 next-token prediction / cross-entropy training。输入序列包含 projected visual tokens 和 language tokens；模型在视觉上下文条件下预测后续文本。关键细节是：cross-entropy loss 通常只在 language tokens 上计算。原因很直接：ground truth values 是与视觉输入相关的文本 tokens，例如 caption、答案或指令响应；visual tokens 来自 vision encoder/projector，并没有自然的 next-token supervision。模型对 visual-token positions 的预测会被 discard 或 mask 掉。这样训练既保留了 LLM 的生成框架，也让视觉 prefix 通过影响 language-token likelihood 学会对齐。

$$\mathcal{L}_{\text{CE}} = - \sum_{t \in \mathcal{T}_{\text{text}}} \log p_{\theta}(y_t | v_{1:m}, y_{<t})$$

考试问法： 如果考“where is VLM cross-entropy computed and why”，写：loss is computed only on language tokens, not visual tokens, because supervision is textual and visual tokens are conditioning context without ground-truth next-token labels。高分补充：这与 text-only LLM loss 相似，但输入前缀包含 projected visual representations。

6. SmolVLM: 为什么小而开放的模型适合当研究样板

Huggingface SmolVLM

- In this lecture, we will focus our attention on [SmolVLM](#), an open source and open weight VLM
- Why should we care about open source and open weight models?
 - “Closed AI Models Make Bad Baselines!” (A. Rogers)
 - This is important to ensure reproducibility and progress
 - Facilitate adaptation and usage in specific downstream application scenarios
- We will study each component of the training pipeline:
 - Reference training dataset
 - Reference model architecture and training regime
 - Evaluation setup

<https://hackingsemantics.xyz/2023/closed-baselines/>

Marafioti, Andrés, et al. "Smolvlm: Redefining small and efficient multimodal models." arXiv preprint arXiv:2504.05299 (2025).

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

20

原 PPT 第 20 页

人话直觉

闭源 frontier model 像一辆封住引擎盖的跑车：你能看见它跑得快，却很难判断是发动机、轮胎、燃油还是赛道让它快。SmolVLM 的价值在于把引擎盖打开：模型权重、训练思路、评测和下游适配都更容易被复现和研究。

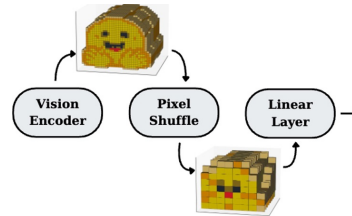
精确定义 / 机制：本讲用 SmolVLM 作为 visual prefix VLM 的 deep dive，因为它是 open source / open weight VLM。讲义引用“Closed AI Models Make Bad Baselines”的观点：闭源模型难以复现、分析和公平比较，不利于科研进步和特定下游场景适配。SmolVLM 的 backbone 思路清晰：用 SigLIP 作为 vision encoder，用 SmolLM2 作为 language backbone，再通过 projector 接入视觉 tokens。它的模型规模包括 256M、500M、2.2B 等变体，目标是在较小显存占用下保持有竞争力的视觉语言能力，尤其适合 edge applications、本地部署和资源受限场景。

考试问法：如果考“why focus on SmolVLM / why open weights matter”，答 reproducibility、progress、adaptation、downstream application。再概括架构：SigLIP vision encoder + SmolLM2 LLM backbone + projector under visual prefix design。

7. Pixel shuffle: 把视觉细节压缩成更少 token

SmolVLM – Pixel Shuffle

- Early methods encode every pixel or patch individually, resulting in lengthy sequences—196 tokens for a 224×224 image at 16×16 resolution
- Compression methods are required to reduce the visual token count and minimise memory consumption
- **Pixel-shuffle**: allows to reduce token counts fourfold while maintaining finegrained visual understanding capabilities (e.g., OCR capability)
- This operation makes sure to reduce spatial resolution and increase representational density (along the depth dimension!)



Marafioti, Andrés, et al. "SmolVLM: Redefining small and efficient multimodal models." arXiv preprint arXiv:2504.05299 (2025).

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

23

原 PPT 第 23 页

人话直觉

Pixel shuffle 像把一张大地图折叠起来：地理信息还在，但不再摊满整张桌子。折得越狠，桌面越省；但如果地图上有很小的街名或门牌号，折太狠就可能看不清。OCR 为什么难，原因就在这里：小字最怕空间细节被压没。

精确定义 / 机制：高分辨率、多图和视频输入会让 visual token count 快速爆炸。早期方法可能为每个 patch 保留 token，例如 224×224、16×16 patch 就有 196 tokens；多张图或多帧视频会线性叠加。SmolVLM 使用 pixel shuffle 一类压缩方法来降低 token 数量和 memory consumption。其核心思想是减少空间分辨率，同时增加 depth dimension 的 representational density：把相邻空间位置的特征重新排列并经 MLP 合并，用更少 tokens 表达更多局部信息。shuffle ratio r 控制压缩和空间保真之间的 trade-off；讲义指出 SmolVLM 使用 $r=4$ 来减少 visual token count，这对小模型尤其关键。

$$N_{\text{after}} \approx \frac{N_{\text{before}}}{r}$$

考试问法：如果考“what does pixel shuffle solve”，回答：it compresses visual tokens to reduce memory/context cost while trying to preserve fine-grained understanding such as OCR。必须讲 trade-off: larger compression saves compute but may lose spatial detail; too little compression preserves detail but keeps cost high。

8. 三阶段训练：先接线，再练读图，最后学视频

SmolVLM – Overview of the Training regime

- SmolVLM is a model able to encode both single image as well as video inputs (i.e., multiple image frames)
- Model training proceeds in **three stages** requiring specialised datasets:
 - **Adaptation stage**: required to teach the projector layer to adapt the vision encoder representations to the language model backbone
 - **Image stage**: required to distil image understanding abilities across a variety of different image formats
 - **Video stage**: required to enable understanding of multi-image content going from single and multiple images, as well video

Marafioti, Andrés, et al. "Smolvim: Redefining small and efficient multimodal models." arXiv preprint arXiv:2504.05299 (2025).

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

27

原 PPT 第 27 页

人话直觉

SmolVLM 的训练不是一锅乱炖，而像课程表。第一阶段是“接线课”：让 projector 学会把视觉信号插进语言模型；第二阶段是“读图课”：重点练 OCR、PDF、图表、表格这些真实图像格式；第三阶段是“连续世界课”：从单张图扩展到多图和视频帧。

精确定义 / 机制： SmolVLM training proceeds in three stages. Adaptation stage 的目标是训练 projector，把 SigLIP vision encoder 的表示适配到 SmolLM2 language backbone；通常保持 language model frozen，并可能对 vision encoder 使用 LoRA，数据多为 noisy web data 和 captioning. Image stage 的目标是提升模型理解不同 image formats 的能力，尤其 OCR、PDF、charts、tables 等；讲义中这一阶段 finetune entire model，包括视觉和 LLM backbone，并加入少量 text-only data 来保留 instruction-following 和减少 catastrophic forgetting. Video stage 进一步让模型理解 multi-image and video content，通过 image → multi-image → video 的 curriculum learning 增加输入复杂度，并使用 image、multi-image、video、text-only 的 careful dataset mixture。

考试问法：如果考“SmolVLM three-stage training”，最稳答法是三行：adaptation = train projector / align visual representations to LLM, LLM frozen；image stage = full finetuning for OCR/documents/charts/tables plus some text to preserve language ability；video stage = curriculum to multi-image/video with mixed datasets。

9. 评测与设计权衡：小显存不等于小问题

Summary

- In today's lecture we saw how to implement frontier Vision and Language models that can be applied to a variety of different tasks
- We analysed the training pipeline for training visual prefix-based VLMs with a focus on SmolVLM, a state-of-the-art Vision and Language model created by Huggingface
 - Trained on open-source data
 - It can be downloaded from Huggingface
 - Thanks to its small memory footprint, it can be used for many difficult application domains
- In visual prefix-based, we saw that the number of tokens allocated for the visual modality has an effect on the accuracy of the models; important trade-off between computational accuracy and accuracy
- Creating Vision and Language models requires careful design of the training pipeline to take advantage of the benefits of pretrained vision and language models without catastrophic forgetting of their initial capabilities

Alessandro Suglia

ATNLP Week 9/ Lecture 26

School of Informatics

40

原 PPT 第 40 页

人话直觉

VLM 设计像装旅行箱：你想带更高清的图、更长的文档、更多视频帧、更大的视觉 encoder、更强的 LLM，但箱子只有那么大。SmolVLM 的故事不是“越小越好”，而是问：在有限 VRAM 和 context budget 下，怎样把视觉、语言、数据和训练策略搭配得最划算？

精确定义 / 机制： SmolVLM 使用 VLMEvalKit 做大规模评测，并强调 computational efficiency：较小变体在单图推理中显存占用很低，适合 edge devices、本地部署、低延迟和隐私敏感场景。评测和 ablation 还揭示几个重要规律：language model capacity 会影响视觉任务，因为 OCR 后理解、指令跟随和答案生成仍依赖语言推理；context length 会影响能容纳的视觉 tokens、长文档和大图；larger vision encoder 的收益在 LLM backbone 也足够大时更明显。未来研究问题包括 fine-grained visual understanding、object-centric representations、多粒度视觉表示、VLM 内部机制，以及“这到底是真正的 vision+language model，还是 language model with some vision？”

考试问法： 如果考“VLM design trade-offs / evaluation insights”，不要只列 benchmark。要写 token count、resolution、context length、vision encoder size、LLM capacity、freeze vs finetune、data mixture、catastrophic forgetting 和 deployment VRAM 之间的权衡。最后可补 future directions：object-centric representations、mechanistic interpretability、real-world multimodal systems。

考试速记版

- Visual prefix VLM: vision encoder 产出 visual tokens，projector 映射到 LLM hidden space，作为 prefix 条件化 LLM 生成。
- 三种融合路线：cross-encoding 把图文一起编码；cross-attention 让文本 attend to vision；visual prefix 把图像变成 prefix tokens。
- Language expert = pretrained LLM；vision expert = pretrained ViT / contrastive encoder；projector = 视觉空间到语言空间的接口。

- ViT patch 公式: $\text{NP}=(H/P)^2$ 。224×224、16×16 patch 得 196 tokens; 384×384、16×16 patch 得 576 tokens。
- 视觉 token 越多, context、显存和 attention 计算越贵; 高分辨率和小 patch 是准确率/细节与成本的 trade-off。
- Contrastive encoders 如 CLIP/SigLIP 用 image-caption pairs 拉近匹配图文、推远不匹配图文, 通常比纯视觉 encoder 更适合接入 LLM。
- Visual prefix training 常用 next-token cross-entropy, 但 loss 只算 language tokens; visual tokens 没有 ground-truth next-token labels。
- Projector 可用 feedforward network 或 cross-attention with learnable queries; 早期可冻结 LLM, 只训练 projector。
- SmolVLM 架构: SigLIP vision encoder + SmolLM2 language backbone + projector, open source/open weight, 适合复现和 edge deployment。
- Image processor 可切 high-resolution image sub-regions; video 可采样 frames, 当作多图输入编码。
- Pixel shuffle 压缩视觉 token: 降低空间分辨率、增加深度表示密度; $r=4$ 代表压缩/保真 trade-off。
- SmolVLM 三阶段: adaptation 学 projector; image stage 强化 OCR/PDF/charts/tables; video stage 扩展到 multi-image/video。
- Image stage 加少量 text-only data 是为了保留 instruction-following, 减少 catastrophic forgetting。
- VLMEvalKit 用于评测; 语言模型容量、context length、视觉 encoder size 都会影响视觉任务表现。
- 未来方向: fine-grained visual understanding、object-centric representations、multi-granularity vision、VLM mechanistic interpretability、real-world multimodal systems。

一段稳妥考场回答

A visual prefix-based VLM reuses two pretrained modality experts: a vision encoder, often a ViT or contrastive image-language encoder such as CLIP or SigLIP, and a text-only LLM backbone. The vision encoder converts an image into visual tokens, for example ViT patch tokens where for a square image $(NP=(H/P)^2)$. These visual representations usually have a different dimensionality and distribution from the LLM hidden space, so a projector maps them into embeddings that the LLM can consume. The projected visual tokens are then prepended as a visual prefix, conditioning the LLM to generate captions, answers or instruction responses. Training usually uses next-token cross-entropy like a text-only LLM, but the loss is computed only on language tokens because the supervision is textual and there are no ground-truth next visual-token labels. SmolVLM is a concrete open weight example: it uses SigLIP as the visual encoder, SmolLM2 as the language backbone and a projector under the visual prefix design. Its training is staged: an adaptation stage aligns the visual encoder to the LLM mainly by training the projector while keeping the language model frozen; an image stage finetunes the model for OCR, documents, charts and tables while mixing in text to preserve language ability; and a video stage uses curriculum learning to extend from single images to multi-image and video inputs. The main design trade-offs are visual token count, image resolution, patch size, projector capacity, freezing versus finetuning, data mixture, context length, VRAM cost and catastrophic forgetting. Pixel shuffle addresses the token bottleneck by reducing spatial resolution while increasing representational density, lowering memory use but risking loss of fine spatial detail such as OCR.